

ECEN 428/722 - Lab Report

Lab Number: 10

Lab Title: Implementation of Advanced Encryption Standard Algorithm

Section Number: 700

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Date: 12/09/2025

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Objectives:

The objective of this lab is to implement and compare the iterative and pipelined architectural variants of the Advanced Encryption Standard (AES) cryptographic algorithm's encryption module. Students will implement one or both versions and then report and compare metrics such as power, area, throughput, and latency for the two architectures. The lab also introduces the use of the Virtual Input Output (VIO) IP core to drive input data (plaintext and key) and read the output data (ciphertext) during runtime on the FPGA.

Methodology:

1. Implement the pipelined and iterative version of the AES encryption module.
2. Instantiate the PLL to generate a 200 MHz clock from the 125 MHz board clock.
3. Instantiate and configure the Virtual Input/Output (VIO) IP core to interface with the AES module.
4. Configure the VIO for 2 input probes (128-bit ciphertext and valid) and 4 output probes (128-bit plaintext, key, and their valid signals).
5. Verify the design using behavioral simulations with specified input vectors.
6. Generate the bitstream and debug file (*.ltx) to test the design onboard the FPGA using the VIO interface.
7. Report and compare the power, area, throughput, and latency metrics for both the pipelined and iterative architectures

Design:

The lab's design centers on implementing the AES Encryption Core within an FPGA top-level file alongside two essential components:

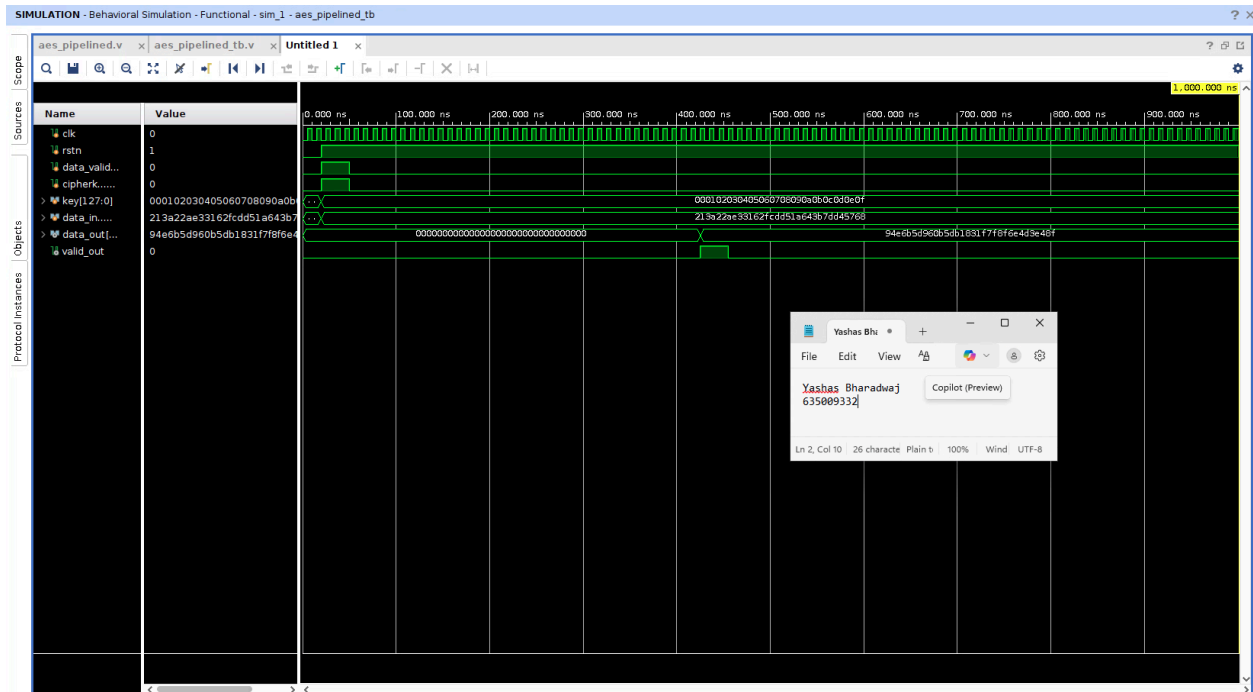
- AES Core: Implements the AES encryption module. Must be designed using the Pipelined architecture.
- PLL (Phase-Locked Loop): Generates the 200 MHz clock required to run the AES core from the FPGA's 125 MHz input clock.
- VIO (Virtual Input/Output) IP Core: Serves as the interface to drive the 128-bit key and 128-bit plaintext into the AES module. Used to read the resulting 128-bit ciphertext from the AES module.

The goal is to implement and compare the physical characteristics of the Pipelined (many round operators used once) and Iterative (one round operator reused 10 times) AES designs.

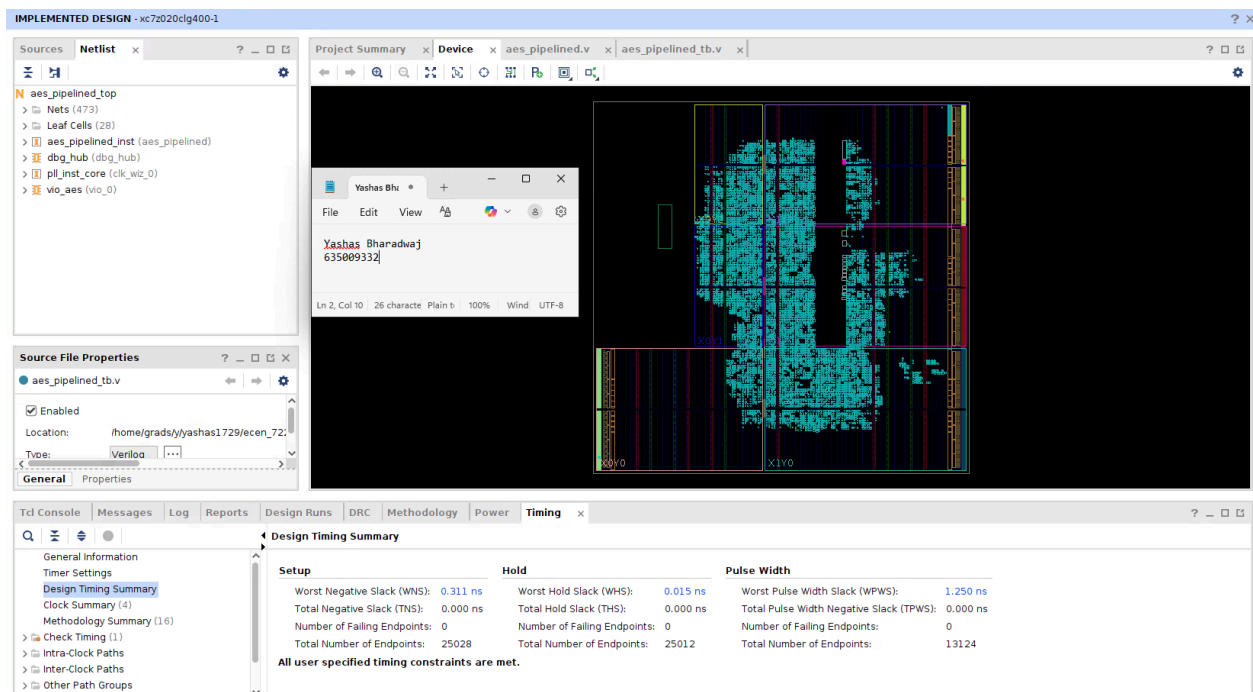
Results:

AES Pipelined

Implemented Design:



Bitstream Generation:



Timing Report:

Tcl Console Messages Log Reports Design Runs DRC Methodology Power Timing x			
Design Timing Summary			
General Information			
Timer Settings			
Design Timing Summary			
Clock Summary (4)			
Methodology Summary (16)			
> Check Timing (1)			
> Intra-Clock Paths			
> Inter-Clock Paths			
> Other Path Groups			
Setup			
Worst Negative Slack (WNS):	0.311 ns	Worst Hold Slack (WHS):	0.015 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	25028	Total Number of Endpoints:	25012
Pulse Width			
Worst Pulse Width Slack (WPWS):	1.250 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Total Number of Endpoints:	13124
All user specified timing constraints are met.			

Since, Total Negative Slack = 0 and Worst Negative Slack > 0, the implemented design is error free and can be successfully implemented on the FPGA.

Power Report:

```
29 1. Summary
30 -----
31
32 +-----+-----+
33 | Total On-Chip Power (W) | 0.436 |
34 | Design Power Budget (W) | Unspecified* |
35 | Power Budget Margin (W) | NA |
36 | Dynamic (W) | 0.326 |
37 | Device Static (W) | 0.109 |
38 | Effective TJA (C/W) | 11.5 |
39 | Max Ambient (C) | 80.0 |
40 | Junction Temperature (C) | 30.0 |
41 | Confidence Level | Medium |
42 | Setting File | --- |
43 | Simulation Activity File | --- |
44 | Design Nets Matched | NA |
45 +-----+-----+
46 * Specify Design Power Budget using, set_operating_conditions -design_power_budget <value in Watts>
47
```

Hardware Utilization/Area:

```
29 1. Slice Logic
30 -----
31
32 +-----+-----+-----+-----+-----+-----+
33 | Site Type | Used | Fixed | Prohibited | Available | Util% |
34 +-----+-----+-----+-----+-----+-----+
35 | Slice LUTs | 10727 | 0 | 0 | 53200 | 20.16 |
36 | LUT as Logic | 10703 | 0 | 0 | 53200 | 20.12 |
37 | LUT as Memory | 24 | 0 | 0 | 17400 | 0.14 |
38 | LUT as Distributed RAM | 24 | 0 | 0 | 0 | 0 |
39 | LUT as Shift Register | 0 | 0 | 0 | 0 | 0 |
40 | Slice Registers | 13069 | 0 | 0 | 106400 | 12.28 |
41 | Register as Flip Flop | 13069 | 0 | 0 | 106400 | 12.28 |
42 | Register as Latch | 0 | 0 | 0 | 106400 | 0.00 |
43 | F7 Muxes | 3232 | 0 | 0 | 26600 | 12.15 |
44 | F8 Muxes | 1600 | 0 | 0 | 13300 | 12.03 |
45 +-----+-----+-----+-----+-----+-----+
46 * Warning! LUT value is adjusted to account for LUT combining.
47
```

From Latency:

According to data_valid_in and data_out signals, it takes 42 clock cycles for the output to be calculated

and delivered.

Therefore, Latency = 42 clock $\Rightarrow 42 * 10\text{ns} = 420\text{ns}$

Throughput of the design:

Stable Throughput is given by the following formula:

Total bits of output generated / Time to generate a new result

But since this is the pipelined version of the AES model, a new result comes every clock cycle.

Therefore, Throughput

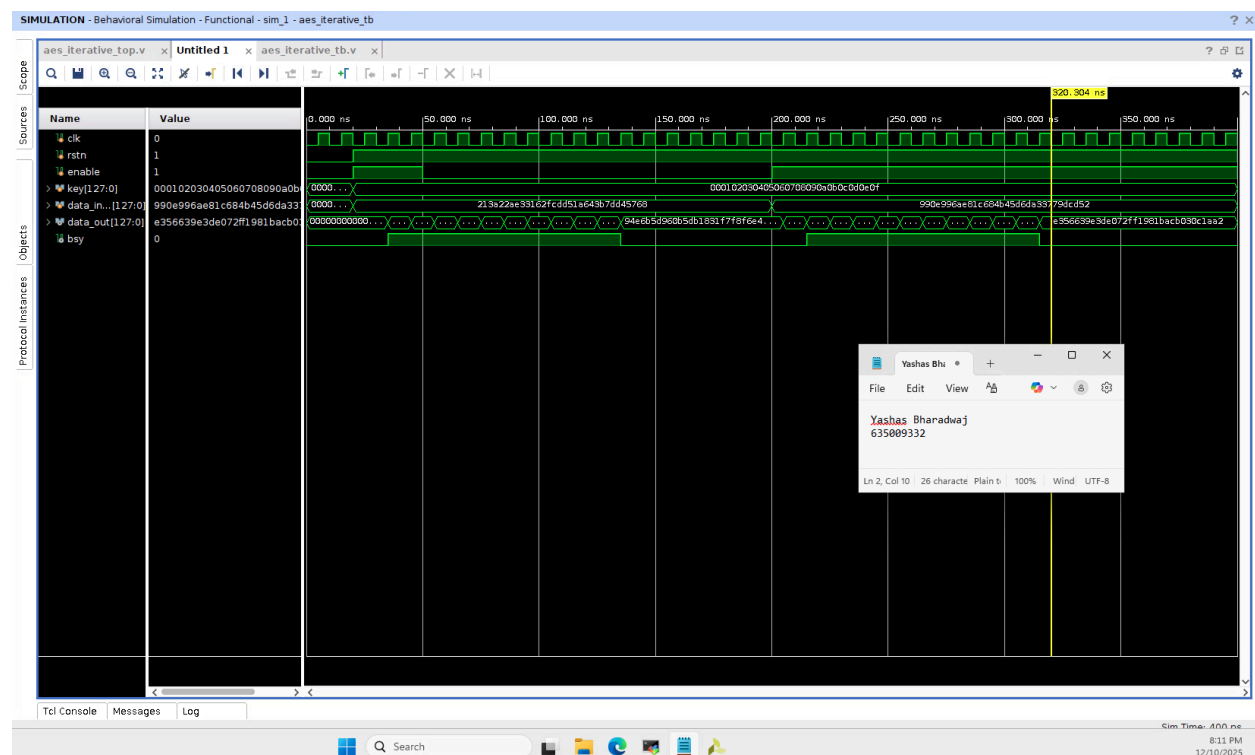
= 128 bits / clock period

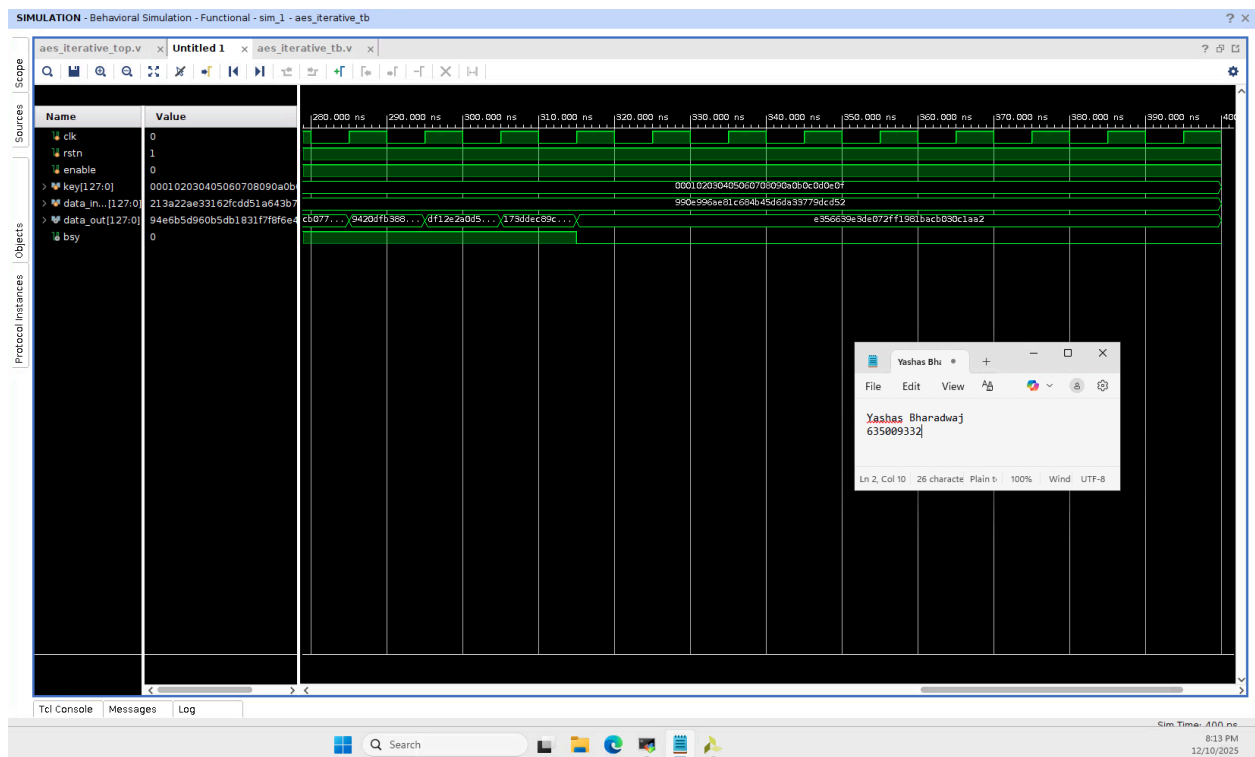
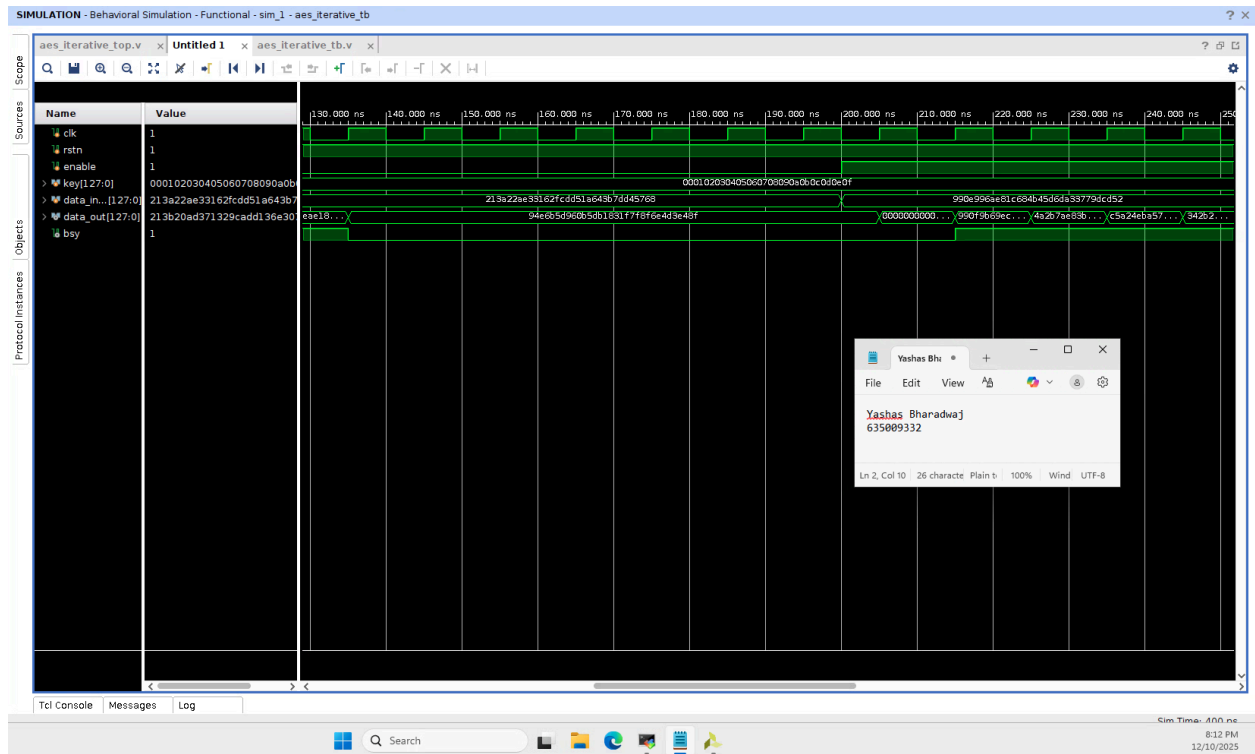
= 128 bits / 10ns

= 1600MBps

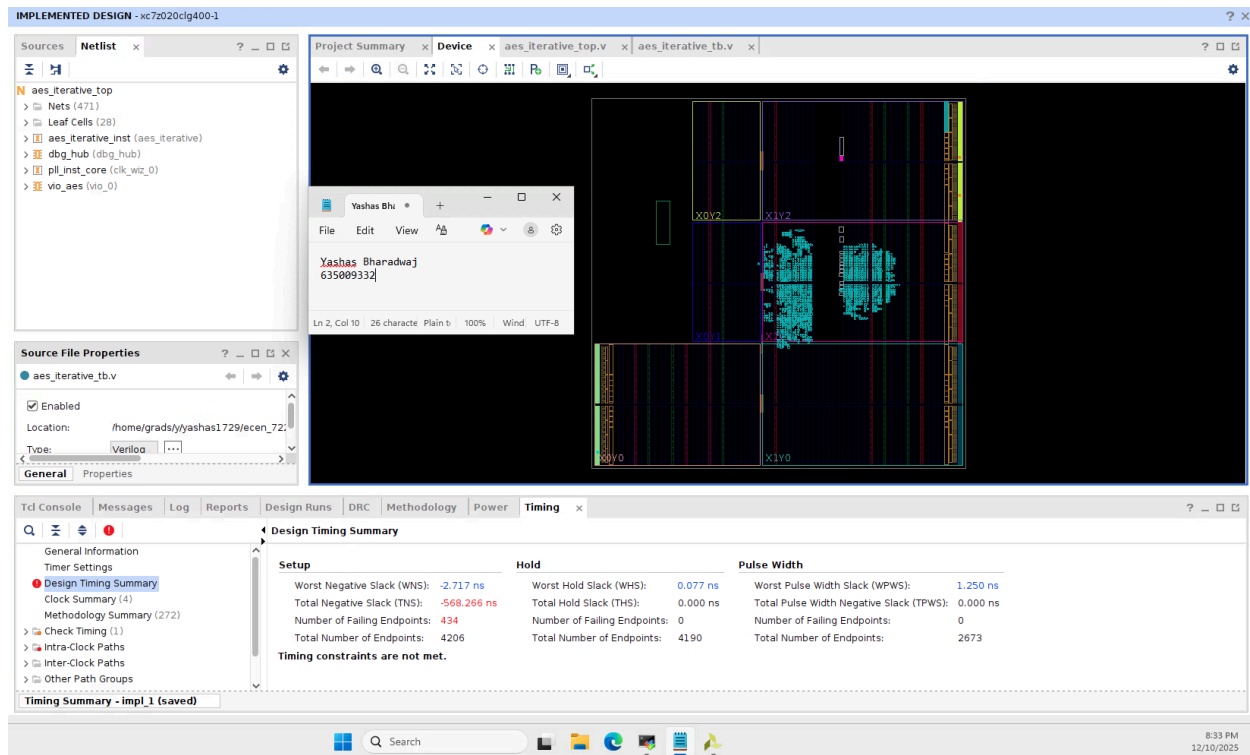
AES Iterative

Behavioural Simulation:

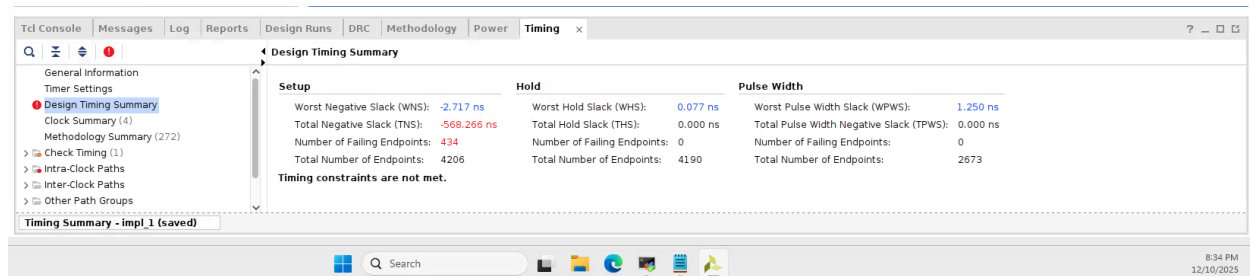




Bitstream Generation:



Timing Report:

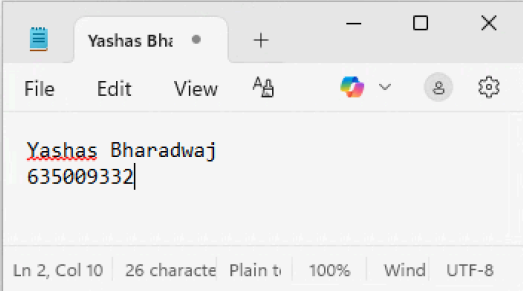


Since, Total Negative Slack $\neq 0$ and Worst Negative Slack < 0 , the implemented design has errors and cannot be implemented on the FPGA.

However, when this design is loaded onto the FPGA, it works perfectly fine. And the result gets displayed on the terminal screen.

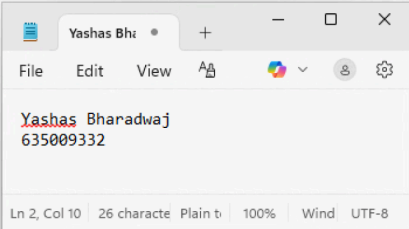
Power Report:

```
29 1. Summary
30 -----
31
32 +-----+-----+
33 | Total On-Chip Power (W) | 0.376 |
34 | Design Power Budget (W) | Unspecified* |
35 | Power Budget Margin (W) | NA |
36 | Dynamic (W) | 0.268 |
37 | Device Static (W) | 0.108 |
38 | Effective TJA (C/W) | 11.5 |
39 | Max Ambient (C) | 80.7 |
40 | Junction Temperature (C) | 29.3 |
41 | Confidence Level | Medium |
42 | Setting File | --- |
43 | Simulation Activity File | --- |
44 | Design Nets Matched | NA |
45 +-----+-----+
46 * Specify Design Power Budget using, set_operating_conditions -design_power_budget <value in Watts>
47
```



Hardware Utilization/Area:

```
29 1. Slice Logic
30 -----
31
32 +-----+-----+-----+-----+-----+-----+
33 | Site Type | Used | Fixed | Prohibited | Available | Util% |
34 +-----+-----+-----+-----+-----+-----+
35 | Slice LUTs | 2819 | 0 | 0 | 53200 | 5.30 |
36 |   LUT as Logic | 2795 | 0 | 0 | 53200 | 5.25 |
37 |   LUT as Memory | 24 | 0 | 0 | 17400 | 0.14 |
38 |   LUT as Distributed RAM | 24 | 0 | 0 | 0 | 0.00 |
39 |   LUT as Shift Register | 0 | 0 | 0 | 0 | 0.00 |
40 | Slice Registers | 2618 | 0 | 0 | 106400 | 2.46 |
41 |   Register as Flip Flop | 2618 | 0 | 0 | 106400 | 2.46 |
42 |   Register as Latch | 0 | 0 | 0 | 106400 | 0.00 |
43 | F7 Muxes | 32 | 0 | 0 | 26600 | 0.12 |
44 | F8 Muxes | 0 | 0 | 0 | 13300 | 0.00 |
45 +-----+-----+-----+-----+-----+-----+
46 * Warning! LUT value is adjusted to account for LUT combining.
47
```



From Latency:

It is given by how much, time it requires to generate a new data

Therefore, Latency = Time when output generated - Time when input was generated

= 135-25ns

= 110 ns

Throughput of the design:

Stable Throughput is given by the following formula:

Total bits of output generated / Time to generate a new result

= 128 bits / 110ns

= 1163 Mbps

= 145 MBps

Comparison of Results:

	Pipelined AES	Iterative AES
Throughput	High Throughput	Low Throughput
Latency	High Latency for each result	Low Latency for each result

Conclusion:

This lab was very informative and we learnt about the implementation of AES Encryption Algorithm which was taught to us in the class. We compared and presented the data with respect to both pipelined and iterative AES algorithms.